Safety

ChatGPT edition — post this at the bench

Roles, boundaries, and the stop-before-touch workflow

Stop line and roles

Learner boundary: isolated battery power only, 12 V DC maximum. No outlets, mains leads, opened appliances, large battery packs, charged power capacitors, ignition coils, sparks, or exposed high voltage. An adult facilitator is not thereby an electrically qualified person.

The learner predicts, assembles only the pictured battery circuit, and records. The adult inspects parts, controls power, and stops heat, odor, damage, pain, or unexpected behavior. Work outside this boundary belongs only to a person qualified for that work.

What you need

- This guide
- Safety glasses
- Parts tray
- Current-limited battery holder with switch

Predict first

An adult is present. Does that make work on an opened mains appliance acceptable? Write the reason, not just yes or no.

LEARNER: battery task | ADULT: inspect / stop | QUALIFIED: other work

hard boundary: roles do not merge

Investigate

1. Work dry, uncluttered, and away from jewelry, loose metal, drinks, and damaged cells.
2. Assemble with power disconnected. Adult checks source, polarity, meter jacks, bare conductors, and likely short paths.
3. Energize only for the stated interval. One hand owns the switch. Disconnect before moving a lead.
4. Stop immediately for heat, swelling, leakage, odor, smoke, pain, damaged insulation, or behavior not predicted.
5. Count small magnets before and after; store them inaccessible to children. Never use a lithium cell in a direct-short demonstration.

Explain from the evidence

Voltage alone is not a safety verdict. Current path and duration affect shock; available current and stored energy affect heating, arcs, and fire. OSHA's qualified-person distinction is a useful boundary even though this is not a workplace course. The course therefore excludes mains and exposed high voltage rather than teaching a learner to manage them.

Change one thing

Run a no-power rehearsal. The learner points to the disconnect, names every stop condition, and demonstrates where each meter lead goes. If the rehearsal is uncertain, do not energize.

Evidence record	
Prediction	______________________
One change	______________________
Observation	______________________
Claim + because	______________________

Physics and safety basis: NIST SI/CODATA; OSHA 29 CFR 1910.333 and OSHA 3075; DOE Electrical Science handbooks; HyperPhysics (Georgia State University); University of Colorado PhET teacher resources. Full citations and scope notes are recorded in the provider-neutral electricity research library.